

Paper 005.5 — Double-Slit Experiment

A Generative Paper of the Event Density Program

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of $\lambda = h/p$, the fringe-spacing geometry $d \sin \theta = m\lambda$, or the visibility-vs-coupling decoherence curve. All three are INHERITED from standard QM + standard wave optics + standard decoherence theory.
2. It does **not** claim a substrate first-principles derivation of the apparatus-specific value of Θ_{ED} (P12 ED-threshold). Substrate derivation in $\ell_{ED} + V1/V5$ kernel parameters is OPEN; apparatus-specific value INHERITED via empirical matching to coherence-loss boundary.
3. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
4. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is **M3 (form-IDENTIFIED + value-INHERITED)** as stated in §1; the audit table contains zero pure-D rows. The structural substrate composition is identified, not forced; the language "form-IDENTIFIED" rather than "form-FORCED" is the §1 / §4 / §6 default.
5. It does **not** introduce new substrate primitives. P-EDThreshold-Collapse is a naming convenience (P12 + P11 + Paper_005 conjunction), not a new postulate. The earlier-draft P-MultiChannel-Capacity and P-WhichPath-as-EDInjection are absorbed into upstream content (Paper_072 individuation; V5 + P12 conjunction respectively).

Abstract

The double-slit experiment is the structural composition of four substrate-level mechanisms acting on a single chain C : (i) **multi-channel participation** below the P12 ED-threshold, with one chain distributing bandwidth across slit-A, slit-B, and downstream channels per P02+P04+P07+Paper_072 — no splitting, no duplication; (ii) **coherent V1 propagation** (Paper_089) preserving the substrate phase difference $\Delta\pi(x)$ across channels; (iii) the **detection-plane participation distribution** $\rho^C(x) = |\sum_K P_K(x)|^2$, whose constructive/destructive structure (substrate phase, Paper_008) gives the fringe pattern as the per-trial commitment-probability density (Paper_003 Born frequency limit + Paper_005 projector); (iv) **threshold-driven collapse** with which-path-as-V5-injection: V5 cross-channel coupling (Paper_063, Paper_090) to a path-distinguishing environment transfers substrate event-density into C , raising local ED above Θ_{ED} and triggering single-channel commitment before the participation field reaches the detector.

Delayed-choice carries no retrocausality: threshold-crossing is local in time and follows V5 coupling at the moment any path-distinguishing apparatus engages the chain. Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED)**. Form IDENTIFIED: existence of the interference pattern as participation distribution; binary structural collapse at

Θ_{ED} ; visibility loss under V5 coupling; locality-in-time of delayed-choice. Value **INHERITED**: fringe spacing, $\lambda = h/p$, Θ_{ED} apparatus-value, visibility-vs-coupling curve. Sharpest falsifier (F1): a quantum carrier whose local ED exceeds Θ_{ED} yet sustains multi-channel interference at the detector — refutes the P12 + P11 + Paper_005 commitment composition.

1 Statement of Result

The double-slit experiment composes four substrate mechanisms on a single chain C :

1. **Multi-channel participation** below Θ_{ED} : $\Psi^C = \sum_K P_K |K\rangle$ with $P_K = \sqrt{b_K} e^{i\pi_K}$ (Paper_001 / Paper_003_5). One chain, multiple channels per P02+P04+P07; chain-identity preserved per Paper_072.
2. **Coherent V1 propagation** (Paper_089) advances the multi-channel amplitudes jointly, preserving $\Delta\pi(x) = \pi_{K_A}(x) - \pi_{K_B}(x)$.
3. **Detection-plane participation distribution** $\rho^C(x) = |\sum_K P_K(x)|^2$ via Born frequency limit (Paper_003) on V1-propagated amplitude. Per-trial: P11 commits to one K^* via Paper_005's projector form. Many-trial: empirical density converges to $\rho^C(x)$.
4. **Threshold collapse and V5 which-path injection.** Crossing Θ_{ED} (P12) triggers irreversible single-channel commitment (P11). Path-distinguishing apparatus couples C to environment E via V5 cross-chain (Paper_063, Paper_090); the V5 coupling injects substrate event-density into C , raising local ED above Θ_{ED} and forcing collapse before V1 propagation reaches the detector. Visibility loss is structural (V5-driven cross-chain ED injection), not epistemic.

Delayed-choice carries no retrocausality: threshold-crossing is local in time and follows V5 coupling at the moment of apparatus engagement.

Form IDENTIFIED: existence of the interference pattern as participation distribution; binary structural collapse at Θ_{ED} ; visibility loss under V5 coupling; locality-in-time of delayed-choice. **Value INHERITED:** fringe spacing $\Delta x = \lambda L/d$ (Born–Wolf path-difference applied to π_K); $\lambda = h/p$ (Paper_012, Paper_012_5; h INHERITED); apparatus-specific value of Θ_{ED} ; visibility-vs-coupling curve.

2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P04 (bandwidth), P07 (channel structure), P09 (U(1)-valued polarity), P11 (commitment-irreversibility), P12 (ED-threshold / stability landscape).

Upstream:

- I-001 / I-003_5: $P_K = \sqrt{b_K} e^{i\pi_K}$ amplitude convention.
- I-003: Born frequency limit $f_K = b_K / \sum b_{K'}$.
- I-005: Projector form Π_{K^*} from P11 commitment-event identification.
- I-008: $\pi_K \in S^1$.
- I-063: Bipartite non-factorizability under V5 cross-channel correlation.
- I-072: Individuation regime — chain-identity preservation across multi-channel participation.
- I-089: V1 finite-width retarded kernel (coherent propagation).
- I-090: V5 cross-chain finite-memory kernel.

- I-WaveOptics: Path-difference geometry (Born–Wolf).
- I-deBroglie: $\lambda = h/p$ (h INHERITED).

Paper-specific naming convention (no new postulate):

- **P-EDThreshold-Collapse:** Naming convenience for the joint action of P12 (threshold exists) + P11 (commitment is irreversible) + Paper_005 (commitment $\leftrightarrow$ projector). Definitional per Paper_095 §2.3 (composition of upstream content under a single name). No new substrate content.

The earlier-draft P-MultiChannel-Capacity and P-WhichPath-as-EDInjection are absorbed into upstream content (Paper_072 individuation regime preserves chain-identity across distributed bandwidth; V5 cross-chain coupling + P12 threshold conjunction supplies the which-path-as-injection identification). Recording the absorption as honest historical accounting; neither remains load-bearing as a paper-specific postulate.

3 Derivation

3.1 Multi-channel participation below threshold

Below Θ_{ED} (P12), a chain C traversing the two-slit aperture has bandwidth distributed across the substrate channels coupled by the aperture geometry:

$$\Psi^C(\text{aperture}) = P_{K_A}|K_A\rangle + P_{K_B}|K_B\rangle, \quad P_{K_\alpha} = \sqrt{b_{K_\alpha}} e^{i\pi K_\alpha}.$$

Per Paper_072, C retains a single substrate identity across this distribution; the multi-channel form is one chain participating in multiple channels, not multiple copies. For symmetric slits, $b_{K_A} = b_{K_B}$.

3.2 Coherent V1 propagation

The V1 kernel (Paper_089) acts on each channel amplitude:

$$P_{K_\alpha}(x) = \int K_{V_1}(x - x'_\alpha) P_{K_\alpha}(x'_\alpha) d^3 x'_\alpha.$$

Linearity of convolution + V1's finite-width retarded support preserves the relative phase $\Delta\pi(x) = \pi_{K_A}(x) - \pi_{K_B}(x)$ in the regime $\ell_{V_1} \ll d$ (slit separation). To leading order in standard far-field geometry:

$$\Delta\pi(x) = \frac{2\pi}{\lambda} \frac{dx}{L}, \quad \lambda = h/p \text{ (INHERITED)}.$$

3.3 Detection-plane participation distribution

Per Paper_003 (substrate-frequency Born identification) the participation density at the detector is the Born rule applied to the V1-propagated multi-channel amplitude:

$$\rho^C(x) = |P_{K_A}(x) + P_{K_B}(x)|^2 = b_{K_A}(x) + b_{K_B}(x) + 2\sqrt{b_{K_A} b_{K_B}} \cos \Delta\pi(x).$$

For symmetric slits with $b_{K_A}(x) \approx b_{K_B}(x) \equiv b_0(x)$:

$$\rho^C(x) = 4b_0(x) \cos^2(\tfrac{1}{2}\Delta\pi(x)).$$

Per-trial: P11 commits to one K^* with probability $\propto \rho^C$ (Paper_005). Many-trial: empirical distribution converges to $\rho^C(x)$ (Paper_003 frequency limit). Bright fringes correspond to $\Delta\pi = 2\pi m$; dark fringes to $\Delta\pi = (2m + 1)\pi$; fringe spacing $\Delta x = \lambda L/d$ from standard wave-optics path-difference geometry applied to the substrate phase π_K .

3.4 Threshold collapse

When local event-density crosses Θ_{ED} (P12), P11 fires once; Paper_005's projector form selects K^* with probability $b_{K^*}/\sum b_{K'}$. Pre-threshold: multi-channel. Post-threshold: $\Psi^C = |K^*\rangle$. The transition is binary, structural, physical — not epistemic. This is the substrate quantum-classical boundary at the per-event level; P-EDThreshold-Collapse is the naming convenience for the P12 + P11 + Paper_005 conjunction.

3.5 Which-path = V5 ED-injection

Path-distinguishing apparatus D at slit α couples to C via V5 cross-chain (Paper_090). The substrate joint state is non-factorizable (Paper_063):

$$\Psi^{CUD} = P_{K_A} |K_A\rangle_C \otimes |D_A\rangle_D + P_{K_B} |K_B\rangle_C \otimes |D_B\rangle_D.$$

Two substrate consequences:

(i) **Strong coupling:** the V5 cross-chain transfers substrate event-density into C 's local ED budget. The substrate-graph derivation of the V5-coupling-to-ED-injection rate from V5 kernel parameters is OPEN; the **identification** that V5 coupling carries substrate ED transfer is what is established here. For sufficient coupling, C 's local ED exceeds Θ_{ED} at the slit; threshold collapse fires immediately and C commits to a single channel before V1 propagation reaches the detector. The interference cross-term is absent; the screen records the single-slit envelope $|P_{K_A}|^2 + |P_{K_B}|^2$.

(ii) **Weak coupling:** the chain's reduced state — obtained by tracing out D — is diagonalized in $\{|K_A\rangle, |K_B\rangle\}$ at the rate set by $\langle D_A | D_B \rangle$. Visibility decreases monotonically with V5 coupling strength; the standard decoherence visibility-vs-coupling curve is INHERITED.

Which-path is V5 cross-channel coupling; it is not knowledge.

3.6 Delayed-choice without retrocausality

Threshold-crossing is local in time: it occurs when cumulative V5 ED transfer into C exceeds Θ_{ED} . If D is inserted before the chain's V1-propagated participation field reaches it, V5 coupling at D 's spatial position injects ED at the time of insertion $\rightarrow$ threshold-crossing $\rightarrow$ single-channel collapse $\rightarrow$ no interference. If D is removed (or absent), no V5 coupling, no ED injection, multi-channel participation persists to the detector $\rightarrow$ interference. The “choice” is the timing of V5 coupling; the substrate response is local, no retrocausality required.

3.7 Channels vs pattern

Channels (P07) exist before the pattern: slit A, slit B, micro-geometric sub-paths, downstream propagation channels $K(x)$. Pattern is the participation distribution $\rho^C(x)$ accumulated over many P11 commitments. Bright fringe = high-strength channel (constructive substrate phase reinforcement); dark fringe = channel where substrate phase cancels.

Mapping: **Channels** $\rightarrow$ **coherent V1 propagation of P_K** $\rightarrow$ **participation distribution $\rho^C(x)$** $\rightarrow$ **per-trial P11 commitment** $\rightarrow$ **many-trial pattern**.

4 Audit Table

#	Step	Label	Notes
1	P02, P04, P07, P09, P11, P12	P	Paper_087.
2	$P_K = \sqrt{b_K} e^{i\pi_K}$	I	Paper_001 / Paper_003_5.
3	Born frequency $f_K = b_K / \sum b_{K'}$	I	Paper_003.
4	P11 commitment $\leftrightarrow$ projector Π_{K^*}	I	Paper_005.
5	Chain-identity preserved across multi-channel participation	I	Paper_072.
6	V1 coherent kernel propagation	I	Paper_089.
7	V5 cross-chain coupling, non-factorizable joint state	I	Paper_063, Paper_090.
8	Multi-channel participation below Θ_{ED}	D-via-I	Composition of P02+P04+P07+P12+I-072 under “one chain across multiple channels.”
9	Phase-preserving V1 propagation in $\ell_{V1} \ll d$ regime	I	Linearity of convolution; carrier identification.
10	Detection-plane density $\rho^C(x) = \sum_K P_K(x) ^2$	D-via-I	Born rule applied to V1-propagated amplitude.
11	Constructive/destructive fringe from $\Delta\pi$	I	Standard wave-optics path-difference.
12	Fringe spacing $\Delta x = \lambda L/d$	I	Born–Wolf, $\lambda = h/p$ INHERITED.
13	Threshold collapse (P-EDThreshold-Collapse)	P	Definitional per Paper_095 §2.3.
14a	V5 coupling $\rightarrow$ ED injection $\rightarrow$ threshold collapse	D-via-I	Composition of V5 + P12 + P11.
14b	Substrate-graph derivation of V5-to-ED-injection rate	OPEN	Identification established; computation not constructed.
15	Reduced-state visibility under weak V5 coupling	I	Standard decoherence theory.
16	Delayed-choice locality-in-time	D-via-I	V5 coupling local in time.
17	Apparatus-specific value of Θ_{ED}	OPEN	INHERITED via empirical matching.
18	Visibility-vs-coupling curve from V5 first principles	OPEN	Standard phenomenology INHERITED.
19	Verdict M3 (form-IDENTIFIED + value-INHERITED)	A $\rightarrow$ position	Per Paper_095 §3.3.

Zero pure-D rows. Audit content matches M3 form-IDENTIFIED + value-INHERITED (composition + identification + standard inheritance, no genuine new substrate-graph derivation). No A rows beyond the verdict-framing row 19.

5 Falsification Criteria

- **F1 (sharpest):** A quantum carrier whose local ED demonstrably exceeds Θ_{ED} (per Paper_012 RB1 substrate-rate identification, with Θ_{ED} pinned by the visibility-loss boundary in the same apparatus class) yet sustains multi-channel interference at the detector — refutes the P12 + P11 + Paper_005 commitment composition.
- **F2:** A path-distinguishing apparatus that produces interference erasure with V5 coupling strength below the threshold required for Θ_{ED} -crossing AND below the visibility-decay

coupling implied by the standard decoherence curve — refutes the V5-driven which-path identification. Operational: visibility decay that exceeds the V5-coupling-implied bound at the same apparatus geometry.

- **F3:** Visibility loss not monotonically increasing in V5 coupling strength — refutes the V5 + P12 + decoherence composition.
- **F4:** A delayed-choice variant in which the future configuration of D affects past commitment statistics in a way not reducible to local-in-time V5-coupling-at-engagement — refutes the locality-in-time of P-EDThreshold-Collapse.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

The ED account replaces wave-particle duality with single-chain multi-channel participation, epistemic collapse with V5-driven substrate ED injection above Θ_{ED} , many-worlds branching with single-chain commitment selection, and retrocausal pilot-wave with local-in-time threshold-crossing. The Channels-vs-Pattern distinction (channels are P07 path geometry existing before the trial; pattern is the participation distribution accumulated over many P11 commitments) is the substrate-side reading of the apparent paradox: there is no “wave” propagating in physical space — there is a single chain whose multi-channel participation amplitude is structured by V1 across the apparatus, with the per-trial commitment-density $\rho^C(x)$ giving the screen pattern in the many-trial limit.

The genuine substrate content is the composition (rows 8, 10, 14a, 16): the participation-distribution-as-pattern identification, the Born-rule application to V1-propagated multi-channel amplitude, the V5-as-which-path-injection identification, and the locality-in-time of delayed-choice. Numerical content (fringe geometry, λ , Θ_{ED} , visibility curve) is INHERITED. Three named OPEN items remain (rows 14b, 17, 18); closing row 14b would consolidate the substrate-graph derivation of which-path mechanism; closing row 17 would consolidate Θ_{ED} as a derived substrate scale.

Verdict: **M3 (form-IDENTIFIED + value-INHERITED)** per Paper_095. Audit table contains zero pure-D rows; the four substantive substrate-side rows (8, 10, 14a, 16) are composition/identification of upstream content, not new substrate-graph derivation. The “form-IDENTIFIED” framing rather than “form-FORCED” is the honest reading: the structural account is identified across the substrate composition, not forced from substrate primitives alone.